

HDRP to URP Converter — User Manual

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Unity Compatibility: 2020.3 LTS and later

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1. Introduction

HDRP to URP Converter is an Editor extension for Unity that automates the conversion of High Definition Render Pipeline (HDRP) materials to Universal Render Pipeline (URP) materials.

When a Unity project is migrated from HDRP to URP, materials using HDRP shaders become broken — they render pink or display as "InternalErrorShader." This tool detects those materials, reads their serialized HDRP properties, and remaps them to the correct URP shader properties in bulk.

Key capabilities:

- Batch convert hundreds of materials in one click
 - Automatically detect broken HDRP materials across the entire project
 - Preserve base color, normal maps, metallic/smoothness values, emissive colors, and transparency modes
 - Non-destructive Save Copy mode
 - Full Undo/Redo support
 - Scene lighting normalization for HDRP physical light units
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2. Requirements

Requirement	Minimum
Unity Editor	2020.3 LTS
Render Pipeline	Universal Render Pipeline (URP)
.NET	.NET Standard 2.0
Platform	Windows / macOS / Linux (Editor only)

Important: The Universal Render Pipeline package must be installed in your project before using this tool. The tool is Editor-only and does not add any runtime overhead to your builds.

3. Installation

3.1 Importing from the Asset Store

1. Purchase or download **HDRP to URP Converter** from the Unity Asset Store.
2. Open the **Package Manager** window: [Window → Package Manager](#).
3. Select **My Assets** from the dropdown.
4. Find **HDRP to URP Converter** and click **Import**.
5. In the Import dialog, ensure all files are checked and click **Import**.

3.2 Verifying Installation

After import, verify that the menu item appears:

Tools → HDRP → URP Converter (Free)

If the menu item does not appear, check the Console window for compilation errors and ensure that the Universal Render Pipeline package is installed.

4. Quick Start

Follow these steps to convert HDRP materials to URP in under one minute.

Step 1. Open the converter window:

Tools → HDRP → URP Converter (Free)

Step 2. Add materials using one of the following methods:

- Click **Find Broken** to automatically detect all broken HDRP materials in the project.
- Click **Add Folder** to select a folder and add all materials inside it.

- Drag and drop materials or folders directly into the drop zone.

Step 3. Choose a Save Mode:

- **Override** — Overwrites the original material files.
- **Save Copy** — Creates converted copies in a **ConvertedMaterials/** subfolder alongside the originals.

Step 4. Click the ► **Convert Materials** → **URP** button.

Step 5. Review the Log panel for results. Each material is marked with:

- ✓ — Successfully converted.
 - ✓ [Unlit] — Converted as an Unlit material.
 - ✗ — Error (error message shown alongside).
-

5. Interface Reference

5.1 Toolbar

The toolbar contains three buttons at the top of the window.

Find Broken

Scans the entire **Assets/** folder for materials whose shader is set to **HDRP/***, **HDRRenderPipeline/***, or **Hidden/InternalErrorShader**. All detected materials are added to the conversion list.

Add Folder

Opens a folder selection dialog. All **.mat** files inside the selected folder and its subfolders are added to the conversion list.

Clear

Removes all materials from the conversion list and clears the log.

5.2 Material Drop Zone

The drop zone is the large area below the toolbar. It accepts:

- Individual **Material** assets dragged from the Project window.
- **Folders** dragged from the Project window (all materials inside are added recursively).

Each material entry in the list has an ✕ button on the right to remove it individually.

The label below the drop zone shows the current count, for example: **14 materials**.

5.3 Save Mode

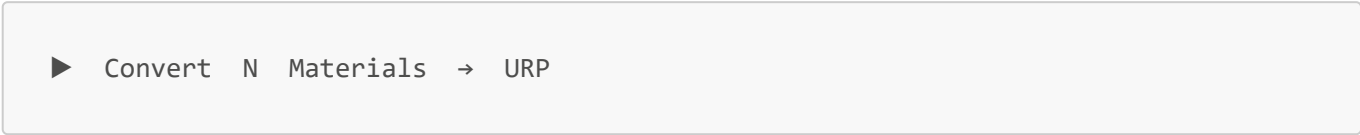
Override

The original material asset is modified directly. Use this when you are confident in the conversion or when working in a version-controlled project.

Save Copy

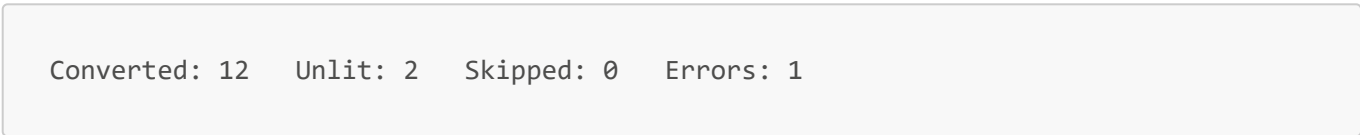
The original material is left untouched. A converted copy is saved in a **ConvertedMaterials/** subfolder within the same directory as the original. If a file with the same name already exists in that folder, a unique name is generated automatically (e.g., **MyMaterial 1.mat**).

5.4 Convert Button



The button is disabled when the material list is empty. Clicking it starts the batch conversion. A progress bar is shown during conversion.

After conversion completes, a summary is shown below the button:



5.5 Log Panel

The log panel appears after the first conversion run. Each line is color-coded:

Color	Meaning
Green (✓)	Material converted successfully
Red (X)	Error occurred; message shown inline
Gray (·)	Skipped (null reference or already URP)

A separator line and summary are appended at the end of each run.

5.6 Scene Lighting Normalization

Click the **Scene Lighting Normalization** section header to expand it.

HDRP uses physically-based light units (Lux for directional lights, Candela for point and spot lights). These values are typically in the thousands and are incompatible with URP's default intensity range of 0–8. This feature patches the **.unity** scene file directly to normalize light intensities.

Usage:

1. Enter a scene path manually, click **Browse** to select a **.unity** file, or click **Active** to use the currently open scene.
2. Click ► **Normalize Scene Lighting**.

What it does:

Action	Detail
Directional lights	Intensity above 10 is set to 1.5
Point and Spot lights	Intensity above 5 is remapped to <code>intensity / 100</code> , clamped to [0.5, 3]
Baked lightmap indices	Reset to 65535 (none)
Lightmap scale offsets	Reset to default (1, 1, 0, 0)
Lighting data asset	Reference cleared
Ambient mode	Set to flat color (0.2, 0.2, 0.22)

Warning: This feature modifies the `.unity` file on disk. Back up your scene or ensure it is under version control before running this feature. Rebake lighting after conversion.

6. Conversion Reference

The following table describes how each HDRP material property is mapped to its URP equivalent.

6.1 Shader Assignment

Condition	URP Shader Assigned
HDRP Lit (default)	<code>Universal Render Pipeline/Lit</code>
HDRP Unlit (shader name contains "Unlit")	<code>Universal Render Pipeline/Unlit</code>

6.2 Property Mapping

HDRP Property	URP Property	Notes
<code>_BaseColor</code> / <code>_Color</code>	<code>_BaseColor</code>	Clamped to [0, 1]
<code>_BaseColorMap</code> / <code>_MainTex</code>	<code>_BaseMap</code>	First non-null texture wins
<code>_NormalMap</code>	<code>_BumpMap</code>	—
<code>_NormalScale</code>	<code>_BumpScale</code>	—
<code>_MaskMap</code>	<code>_MetallicGlossMap</code> + <code>_OcclusionMap</code>	Same texture assigned to both
<code>_MetallicRemapMin/Max</code>	<code>_Metallic</code>	Lerped across remap range
<code>_SmoothnessRemapMin/Max</code>	<code>_Smoothness</code>	Lerped across remap range
<code>_EmissiveColor</code> (HDR)	<code>_EmissionColor</code>	Normalized to LDR if max channel > 1
<code>_EmissiveColorLDR</code>	<code>_EmissionColor</code>	Used when present
<code>_EmissiveColorMap</code>	<code>_EmissionMap</code>	—

HDRP Property	URP Property	Notes
<code>_AlphaCutoff</code>	<code>_Cutoff</code>	—
<code>_SurfaceType == 1</code>	<code>_Surface = 1</code> (Transparent)	—
<code>_AlphaCutoffEnable == 1</code>	<code>_AlphaClip = 1, _ALPHATEST_ON</code>	—

6.3 Keyword Mapping

Condition	Keywords Enabled
Normal map present	<code>_NORMALMAP</code>
Mask map present	<code>_METALLICSPECGLOSSMAP</code>
Emission present	<code>_EMISSION</code>
Transparent surface	<code>_ALPHAPREMULTIPLY_ON</code>
Cutout surface	<code>_ALPHATEST_ON</code>

7. Demo Scene

A demo scene is included to help you explore the converter with sample assets.

7.1 Creating the Demo Scene

1. In the Unity menu, go to:

```
Tools → HDRP → URP Converter → Create Demo Scene
```

2. The tool creates the scene and sample materials at:

```
Assets/Ahuku/HDRP2URP/Demo/HDRP2URP_Demo.unity
Assets/Ahuku/HDRP2URP/Demo/Materials/
```

3. A dialog confirms the creation and shows next steps.

7.2 Using the Demo Scene

- 1. Open `HDRP2URP_Demo.unity`.
- 2. Open the converter: `Tools → HDRP → URP Converter (Free)`.
- 3. Click **Add Folder** and select `Assets/Ahuku/HDRP2URP/Demo/Materials`.
- 4. Click ► **Convert Materials → URP**.
- 5. Observe the log and see the materials updated in the scene.

8. Known Limitations

1. **MaskMap channel packing:** HDRP packs four channels into MaskMap (R=Metallic, G=AO, B=Detail mask, A=Smoothness). URP uses separate textures for Metallic (`_MetallicGlossMap`, R+A channels) and Occlusion (`_OcclusionMap`, G channel). The same MaskMap texture is assigned to both URP slots; results may require manual adjustment for best quality.
 2. **Detail maps:** HDRP detail maps (`_DetailMap`) are not converted. Detail features must be set up manually in URP.
 3. **Subsurface scattering:** HDRP subsurface scattering and translucency properties are not transferred.
 4. **Decal and Hair shaders:** HDRP/Decal and HDRP/Hair shaders are not supported.
 5. **Scene lighting normalization:** The intensity values used for normalization (1.5 for directional, 0.5–3 for point/spot) are heuristic defaults. Fine-tuning may be required depending on your scene's artistic intent.
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9. Frequently Asked Questions

Q: My materials still appear pink after conversion. What went wrong?

A: Check the Log panel for error messages. The most common cause is that the Universal Render Pipeline package is not installed. Go to **Window → Package Manager**, find URP, and install it.

Q: Will this tool modify my original materials?

A: Only if you use **Override** mode. In **Save Copy** mode, original materials are never touched.

Q: Can I undo a conversion?

A: Yes. The tool uses Unity's Undo system. Press **Ctrl+Z / Cmd+Z** immediately after conversion to revert changes.

Q: The Find Broken button found 0 materials, but I know HDRP materials exist.

A: Find Broken searches for materials with shader names containing `HDRP/`, `HDRRenderPipeline`, or `InternalErrorShader`. If your HDRP materials use a custom or renamed shader, use **Add Folder** to add them manually.

Q: Is this tool safe to use on a live project?

A: We recommend using **Save Copy** mode and ensuring your project is backed up or under version control before running batch conversion.

10. Support

If you encounter an issue not covered in this manual, please contact support:

- **Publisher:** Ahuku
- **Email:** [your support email]
- **Unity Asset Store page:** [your asset store URL]

Please include your Unity version, URP package version, and the full text from the Log panel when reporting issues.